RLHF

Reinforcement Learning from Human Feedback

↳ Training LLMs on human preferences

Feedback used in parameter update

Prompt → $\boxed{\text{LLM}}$ → Response → Liked / Disliked

But it is a slow process. So, OpenAI replaced human with a reward model, which was trained on human preferences.

Reward model

↳ Proxy for human preferences

Prompt → $\boxed{\text{LLM}}$ → Response 1, Response 2, Response 3

Ranked

→ Response 3

→ Response 1

→ Response 2

Giving rating is more complex than ranking. Based on some guidelines humans ranked the generated outputs from the same prompt.

Pair-wise preference dataset is created from the rankings & reward mode is trained.

How to update the parameters?

Proximal Policy Optimization (PPO) → Will study later

One big limitation with RLHF:

The trained model will be only as good as reward model & the reward model is only as good as preference data.

DPO

Direct Policy Optimization

↳ Reformulating RLHF as a SL problem

How it works?

First we need a preference dataset

Prompt	Winer	Looser

Prompt → LLM → Probability of prefered response

→ Probability of disprefered response

So we can update the models parameter to increase the probability of generating the prefered response & decrease the probability of generating disprefered response.